

Poland

Office of Competition and Consumer Protection

As the digital marketplace continues to evolve, exploitative methods targeting consumer behavior evolve alongside it. Among the most pressing challenges today are dark patterns—deceptive design techniques that manipulate user choices online, often to the detriment of consumer welfare. Despite growing awareness, practical enforcement standards for addressing dark patterns remain underdeveloped. Recognizing this regulatory gap, the Polish Office of Competition and Consumer Protection (UOKiK) has launched a three-year, EU-funded initiative to design a methodological framework for proceedings involving dark patterns and to explore the application of artificial intelligence (AI) in their detection.

The project, running from May 2023 to April 2026, aims to integrate insights from consumer psychology, digital marketing, behavioral economics, computer science, and legal informatics. At its core, the initiative seeks to develop a Proof of Concept (PoC) for an AI-powered dark patterns detection tool, while also establishing broader guidelines for regulatory use of AI in consumer protection.

I. Building the Foundation: Data Collection and Methodology Development

In the project's first phase, UOKiK's team conducted an internet sweep, reviewing 316 websites—primarily in the e-commerce sector—to identify common dark pattern techniques. These findings provided the first training datasets for the envisioned AI model. Parallel efforts included a review of consumer complaints submitted to UOKiK, helping to prioritize sectors where dark patterns were most prevalent.

To deepen the analysis, neuromarketing experiments were conducted to study consumer responses when exposed to manipulative online interfaces. These tests employed:

- Eye Tracking (ET) to monitor visual attention;
- Facial Tracking (FT) to analyze emotional responses;
- Electroencephalography (EEG) to measure neural activity.

The results revealed that manipulative interface designs significantly influence user decision-making—often without the consumer’s awareness—underscoring the urgency of regulatory intervention.

II. Behavioral Research and Technical Analysis

To validate the neuromarketing findings at scale, UOKiK organized a quantitative online survey simulating real-world shopping scenarios. Results confirmed that dark patterns affect users regardless of education or income level, reinforcing concerns raised in the European Commission’s Fitness Check of EU Consumer Law on Digital Fairness (Brussels, 3 October 2024), which cited dark patterns as one of the most problematic areas in digital consumer protection.

Simultaneously, technical experts analyzed the source code of 100 websites flagged for dark patterns. Code segments responsible for manipulative interface elements were extracted to serve as additional training data, offering a machine-readable interpretation of deceptive design.

III. Leveraging AI: Customizing Large Language Models

In its exploration of AI solutions, UOKiK adapted a strategy inspired by Stuart Mills and Richard Whittle, experimenting with OpenAI’s GPT-4.¹³ While the base model demonstrates advanced comprehension, it is not inherently tailored for regulatory tasks. To address this, UOKiK’s team developed a customized version of GPT-4, fine-tuned to detect dark patterns based on specific guidelines.

Comparative testing between human analysts and the AI model showed high consistency, with GPT-4 correctly identifying patterns found in the manual review—and occasionally detecting additional instances overlooked by humans. Its ability to assess every website component with unbiased precision, free from fatigue or cognitive bias, highlighted its potential as a regulatory support tool.

IV. Next Steps: From Prototype to Practice

Findings from these pilot tests will inform the forthcoming PoC of the AI dark patterns detector, which will combine website structural analysis, neuromarketing and behavioral insights, and source code review.

¹³ Stuart Mills & Richard Whittle, *Detecting Dark Patterns Using Generative AI: Some Preliminary Results* (2023) (unpublished manuscript) (on file with author).

The system will be deployed via a custom-built API to facilitate integration without requiring full-scale tool development—ensuring cost-effectiveness and scalability for regulators.

In February 2026, UOKiK will present the completed PoC and project findings at a closing conference. The deliverables will include not only the AI tool prototype, but also a comprehensive methodology for conducting regulatory proceedings involving dark patterns.

UOKiK's prior work in the AI domain is documented in a White Paper on AI for consumer protection, which provides actionable recommendations for integrating AI into regulatory workflows.¹⁴ Together, these initiatives demonstrate a practical, data-driven pathway for empowering competition and consumer agencies to keep pace with the evolving digital landscape.

¹⁴ UOKiK, *White Paper: Advancing Consumer Law Enforcement with Artificial Intelligence* (2024), <https://uokik.gov.pl/publikacje>.